# Venkata Lakshmi Parimala Pasupuleti

Risk Analysis Lead · Multifamily Affordable Housing · LIHTC · Section 8 · HCV · HUD · Financial Analytics · Compliance · Python · SQL

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Virginia, USA — Open to DC / Hybrid

## PROFESSIONAL SUMMARY

Risk Analysis professional with 8+ years managing multifamily affordable housing compliance, financial risk controls, and lender-facing regulatory workflows across **Section 8/HCV, LIHTC, and HUD** portfolios — the same federal guidelines Fannie Mae's MF lender ecosystem operates under. **Designed and implemented compliance controls** covering 100K+ units with zero violations: AMI threshold calculations, income eligibility verification, recertification governance, and escalation routing. Built **production AI pipeline processing 1.6M financial records** (Python, PyTorch) automating workflows previously handled manually. Applied **anomaly detection (Isolation Forest)** across 3 research pipelines; built quantitative models achieving **$R^2 = 0.977$**. Designed strategic dashboards improving decision-making comprehension **40%** (Tableau). Strong AI curiosity — built agentic compliance audit system (LeaseI) achieving **93.6% precision across 10 compliance rules**. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

## WORK EXPERIENCE

### Entrata
Jan 2017 – Aug 2024

**Senior Software Engineer — Multifamily Risk Management · Affordable Housing Compliance · Financial Analytics**

- **Designed and implemented risk controls and compliance monitoring** for Section 8/HCV, LIHTC, and HUD multifamily portfolios — AMI threshold calculations, income eligibility and credit verification, recertification governance workflows, and escalation routing across 100K+ unit portfolios; **zero compliance violations** over 7+ years; collaborated cross-functionally with compliance, operations, legal, and product stakeholders to identify control gaps and remediate risks.
- **Led financial analysis** for affordable housing programs: HAP allocation, rent pricing strategy (gross rent updates, AMI-based pricing tiers), tax credit calculations, HUD payments, rental subsidies, ledger management, and payment processing — analytical, quantitative risk analysis ensuring adherence to federal and state consumer housing regulations; supported governance and approval processes for eligibility determinations across 100K+ unit portfolios.
- Built **production AI pipeline processing 1.6M financial records** (Python, PyTorch, OpenAI API) automating manual Excel workflows — output validation, quality gates, and compliance controls at launch; eliminated **80% of manual operational workload**; optimized **200+ SQL queries** (30% performance gain); mentored 5+ engineers; achieved **100% PHI compliance** on AWS S3 migration with zero security incidents.

### George Mason University
Aug 2024 – Aug 2025

**Research Associate — Quantitative Risk Analytics · Financial Modeling · Stakeholder Reporting**

- Built multi-source quantitative financial models (Python, R) achieving **$R^2 = 0.977$** across 190+ countries — data collection, hypothesis testing, trend identification, and validation pipeline; delivered risk analysis and strategy recommendations to senior faculty stakeholders.
- Surfaced financial risk anomalies manual review missed across 3 research pipelines by applying **Isolation Forest** — identified control gaps and proposed corrective actions for downstream model reliability.
- Designed strategic risk dashboards (Tableau, ggplot2, Quarto) improving decision-making comprehension **40%** across two cohorts; mentored **600+ students** in data analysis, Python, and database design as Graduate Teaching Assistant.

## PROJECTS

### LeaseI: AI Compliance Audit Pipeline for Affordable Housing — 93.6% Precision
Python · LangGraph · Scikit-learn · SQLite · Deterministic Rules Engine · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built an end-to-end AI compliance audit pipeline for affordable housing lease documents (1,240 files, 5 US states + federal standards, LIHTC/Section 42) — deterministic rule-based engine with full data lineage; achieved **93.6% precision, 87.9% F1** across 10 compliance rules; every determination traceable and audit-ready. Directly mirrors Fannie Mae's compliance monitoring and control framework.

### CrashClarity: Quantitative Risk Analytics on 104K Records
Python · R · Scikit-learn · Random Forest · Logistic Regression · Feature Engineering · Tableau — Live Dashboard

- Mined 104K+ records — built Random Forest + Logistic Regression risk classification model with hypothesis testing achieving **85%+ accuracy**; deployed self-service Tableau dashboard translating quantitative risk outputs into actionable insights for non-technical senior stakeholders.

## EDUCATION

**M.S. Data Analytics Engineering** — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · Quantitative risk modeling, financial analytics, compliance analytics, anomaly detection, data governance, stakeholder reporting

**B.Tech Computer Science & Engineering** — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017

## TECHNICAL SKILLS

**Risk Management & Compliance:** Multifamily Affordable Housing Risk (LIHTC, Section 8/HCV, HUD, AMI), Compliance Controls Design, Control Gap Identification, Risk Mitigation Strategy, Impact Assessments, Income Eligibility & Credit Verification, Recertification Governance, Escalation Routing, PHI/HIPAA Compliance, Audit-Ready Pipelines, Financial Statement Analysis, Lender Process Evaluation

**Analytics & Modeling:** Python (expert), R, SQL (expert — 200+ production optimizations), PL/SQL; Quantitative Risk Modeling, Hypothesis Testing, Regression Analysis, Isolation Forest (Anomaly Detection), Random Forest, Logistic Regression, Feature Engineering, Scikit-learn, Pandas, NumPy; Tableau, Power BI, ggplot2, Quarto, Excel (advanced)

**AI & Cloud Infrastructure:** AI in financial workflows (LangGraph, RAG pipelines, OpenAI API, PyTorch), AWS (S3, EC2, RDS, Glue), ETL/ELT Pipelines, PostgreSQL, MySQL, Docker, GitHub Actions, Salesforce (CRM integrations), Git, Agile/Scrum (120+ sprints)